

MAN00035C

Quantitative Analysis in Business Studies

2026 – Week 3



UNIVERSITY
of York

School for Business
and Society



Week 3: Probability Distributions

Recall from the previous lecture

Combinations and Permutations

- Problems frequently exist, where selections are made from groups of items. In these cases, it is useful to be able to calculate the number of selections that are possible,
 - If the **order** of selection is **important**, we use **permutations**
 - If the **order** of selection **does not matter**, we use **combinations**

Permutations

- With replacement/repetition – e.g. pin code

$$P_{(n,r)} = n^r$$

- Without repetition – e.g. bookshelf

$${}_n P_r = \frac{n!}{(n-r)!}$$



[This Photo](#)



[This Photo](#)

Permutations without repetition

Imagine you have 10 different books and want to calculate how many possible ways you can arrange them on a bookshelf which has a capacity for 5 books. After you place the first book, the second book must be a different book.

$$\begin{aligned}
 {}^n P_r &= \frac{10!}{(10-5)!} \\
 &= \frac{10!}{5!} = \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{5 \times 4 \times 3 \times 2 \times 1} \\
 &= 10 \times 9 \times 8 \times 7 \times 6 = 30,240
 \end{aligned}$$

Check if you can calculate it on your calculator.



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Combinations

- With repetition – e.g. ice cream shop

- $$C_{(n,r)} = \frac{(r+n-1)!}{r!(n-1)!}$$



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- Without repetition – e.g. lottery

- $${}^nC_r = \frac{n!}{r!(n-r)!}$$



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Combinations without repetition

What is the probability of winning the National Lottery (Lotto) jackpot?

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

$$\begin{aligned} {}^{49}C_6 &= \frac{49!}{r!(n-r)!} = \frac{49!}{6! \times (49-6)!} = \frac{49 \times 48 \times 47 \times 46 \times 44 \times 43!}{6! \times 43!} \\ &= \frac{49 \times 48 \times 47 \times 46 \times 44}{6!} = 13,983,816 \end{aligned}$$

Combinations and Permutations

- Summary table

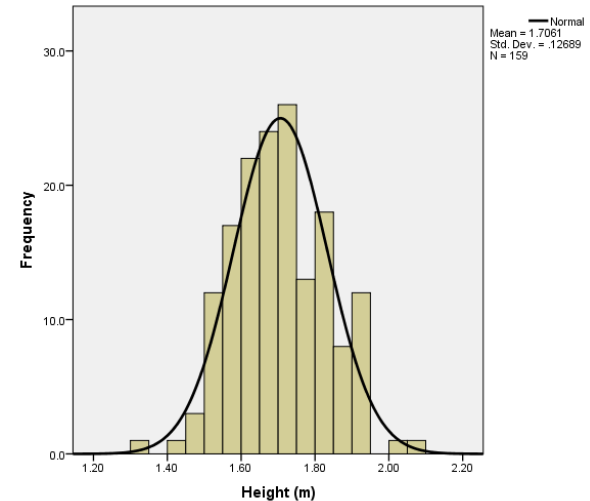
	Order matters Permutations	Order doesn't matter Combinations
No repetition/replacement	${}_nP_r = \frac{n!}{(n-r)!}$	${}_nC_r = \frac{n!}{r!(n-r)!}$
With repetition/replacement	$P_{(n,r)} = n^r$	$C_{(n,r)} = \frac{(r+n-1)!}{r!(n-1)!}$

Session plan

■ Probability distributions:

- Bernoulli
- Binomial
- Normal

Reading: Oakshott, chapter 7 and Taylor, chapter 5.



Bernoulli

Let's start with a **Bernoulli trial**.

It is used to model situations where there are only **two outcomes** (dichotomous variables).

We can give the outcome we “want” the value of **1**, and the other, **0**.

Examples: yes or no; true or false



Coin Flips

- In essence, the **Bernoulli trial (process)** models (weighted) coin flips.
- All we need is the **probability of the outcome** we want, which is exactly the **proportion** of that outcome in the population.
- We say that *the **Bernoulli** is parameterised by p* , the probability of the outcome we want.



		Gender			
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Male	64	40.3	40.3	40.3
	Female	95	59.7	59.7	100.0
	Total	159	100.0	100.0	

Bernoulli trial

That is, we have two possible values for a random Bernoulli trial:

- $X = 1$, with a probability of p
- $X = 0$, with a probability of $1-p$

Note that:

- Only one of these two events can happen,
- and one of the two must happen;
 - They are **mutually exclusive**,
 - and they are **collectively exhaustive**; hence probabilities add to 1.



Bernoulli distribution

If a random variable X has a **Bernoulli distribution** with probability of success p , then

- the **mean** is $E(X)=p$
- with **variance**, $\text{Var}(X) = \sigma^2 = p(1-p)$
- and **standard deviation** $\sigma = \sqrt{p(1-p)}$

Bernoulli – example 2

Suppose that I'm interested if **tomorrow** financial markets go **up** or **down**

- One experiment
- Two possible outcomes

Hence, I can model the set of possible outcomes with a Bernoulli distribution

The **mean** of a Bernoulli with probability p is p .

i.e., if X follows a Bernoulli, $E(X)=p$

with **variance**, $\text{Var}(X) = \sigma^2 = p(1-p)$

and **standard deviation** $\sigma = \sqrt{p(1-p)}$

Bernoulli – example 2

Suppose that I have a portfolio of financial assets and I'm interested if **tomorrow** financial markets go **up** or **down**. We have:

- One experiment
- Two possible outcomes

Hence, I can model the set of possible outcomes with a Bernoulli distribution

My portfolio will make a **profit** if tomorrow financial markets go **up**.

I call the event “tomorrow financial markets go up” a **success**

I allocate the **probability of a success** to the Bernoulli parameter **p**

Bernoulli – example 2

Suppose that based on **past observations** I infer that the **probability of a success**, meaning the **probability that tomorrow financial markets go up** is 0.6

Hence, the (single) **parameter** of the Bernoulli distribution is **$p = 0.6$**

The **mean** is then **$p=0.6$** . Financial markets go up on average 60% of the days

The **variance** is $p(1-p) = 0.6 \times (1-0.6) = 0.24$

The **standard deviation** is $\sqrt{p(1-p)} = \sqrt{0.24} \approx 0.489$

(measure of dispersion around the mean)

Binomial distribution

- A sequence of n repeated (same probability), independent **Bernoulli trials**
- Over n people (*or things*), we basically count **how many** have (*or are, or did*) the outcome we are looking for.
- How many **successes** occur in a sequence of Bernoulli trials
- That is literally just **adding n Bernoulli trials**, assuming they all have the same p , and are **independent**.
- That's something we can work with!

Binomial Distribution

- Adding n (*independent*) Bernoulli trials gets us the **Binomial** Distribution.
- The equations on how to combine expectations and variances from previous lectures apply, if X has a Binomial distribution, then:

$$E(X) = np$$

and

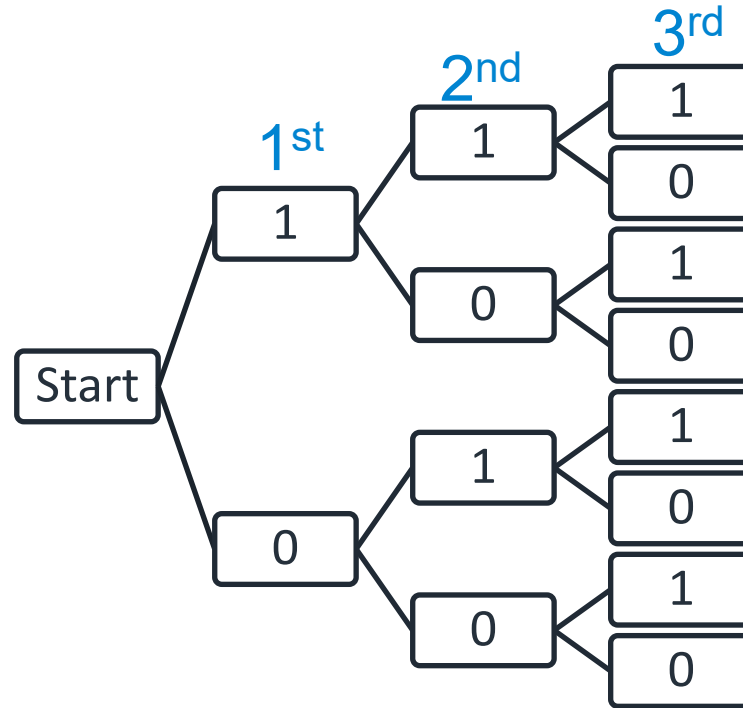
$$\text{Var}(X) = np(1 - p)$$

Note n and p determine the **shape** of the **binomial distribution**; it is *parameterised* by them.



The Binomial Distribution

- These **measures (moments) of the distribution** refer to what happens on average.



The Binomial Distribution

- If we are asking n people a question, and we know the proportion, p , of people who say **yes** then we can calculate **the probability that exactly r of the n say yes.**
- Example:
 - We ask 3 people (independently) to answer **yes** or **no** to a question:
sequence of $n=3$ Bernoulli trials
 - Each person answers yes with probability p : probability of success is p
 - What is the probability that one (two or three) person say yes?

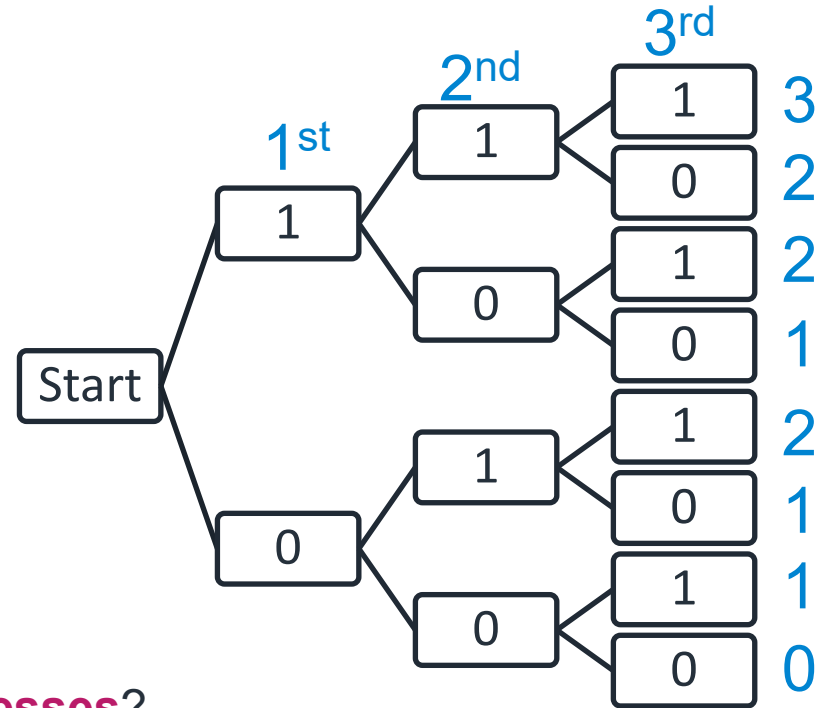
The Binomial Distribution

n is the **number of trials**,

r is the **number of “successes”**

p is the **probability of the result you’re looking for or probability of a success.**

- What are the possible sequences?
- How many branches give r **successes**?
- What’s **the probability of getting r successes**?



The Binomial Distribution

How many branches give r successes?

- r out of n people say yes, so nC_r branches.
- The number of possible sequences/branches is the number of possible ways of choosing r slots for successes out of n trials

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

- Excel: COMBIN(number,number_chosen)

The Binomial Distribution

- What is the probability of each of those branches?

Each of the r successes comes with a probability of p , each of the $n - r$ failures (not-successes) has a probability of $1 - p$.

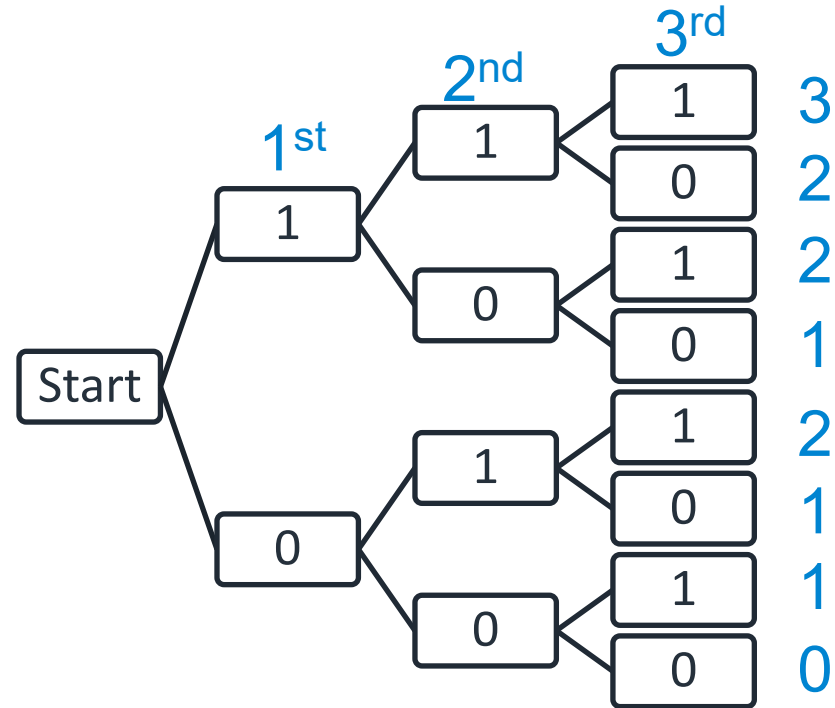
So each branch must have a probability of $p^r (1 - p)^{n-r}$

(they are independent, so we can just multiply probabilities)

The Binomial Distribution

- r out of n people say yes, so nC_r branches
- And each branch must have a probability of $p^r(1-p)^{n-r}$

(they are independent events, so we can just multiply probabilities!)



The Binomial Distribution

In general, the formula to use is:

$$P(\text{r successes in n trials}) = {}^nC_r \times p^r \times (1 - p)^{n-r}$$

(Literally, the number of branches times the probability of each branch.)

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

Where:

n is the **number of trials**,

r is the **number of “successes”**, and

p is the **probability of observing/obtaining a success**

Example

The probability that an invoice will be returned because of an error is **0.1**. If there are **20 invoices** what is the probability that

(a) exactly 2 invoices will be returned?

$$P(r \text{ successes in } n \text{ trials}) = {}^nC_r \times p^r \times (1 - p)^{n-r}$$

Probability of exactly two invoices returned in error

$${}^{20}C_2 = 190$$

$$P(r = 2) = 190 \times 0.1^2 \times 0.9^{18} = 0.2852$$

Example

The probability that an invoice will be returned because of an error is 0.1. If there are **20 invoices** what is the probability that

(b) more than 2 invoices will be returned

$$P(r \text{ successes in } n \text{ trials}) = {}^nC_r \times p^r \times (1 - p)^{n-r}$$

$$P(r > 2) = 1 - [P(r = 0) + P(r = 1) + P(r = 2)]$$

$$P(r = 0) = 1 \times .1^0 \times .9^{20} = 0.1216$$

$$P(r = 1) = 20 \times .1^1 \times .9^{19} = 0.2702$$

$$P(r = 2) = 190 \times .1^2 \times .9^{18} = 0.2852$$

$$P(r > 2) = 1 - (.1216 + .2702 + .2852) = 1 - 0.677 = 0.323$$

EXCEL: BINOM.DIST function

- Returns the individual term binomial distribution probability.
- Use **BINOM.DIST** in problems with a fixed number of tests or trials, when the outcomes of any trial are only success or failure, when trials are independent, and when the probability of success is constant throughout the experiment.
- For example, **BINOM.DIST** can calculate the probability that two of the next three babies born are male.
- Source: Help facility in Excel

EXCEL: The BINOM.DIST function syntax has the following arguments:



- **BINOM.DIST(number_s, trials, probability_s, cumulative)**
- **Number_s** Required. The number of successes in trials.
- **Trials** Required. The number of independent trials.
- **Probability_s** Required. The probability of success on each trial.
- **Cumulative** Required. A logical value that determines the form of the function. If cumulative is TRUE, then BINOM.DIST returns the cumulative distribution function, which is the probability that there are at most number_s successes; if FALSE, it returns the probability mass function, which is the probability that there are number_s successes.

Back at the example

The probability that an invoice will be returned because of an error is **0.1**. If there are **20 invoices** what is the probability that

(a) Probability of exactly 2 invoices will be returned?

$$P(2 \text{ successes in } 20 \text{ trials}) = {}^{20}C_2 \times p^2 \times (1 - p)^{20-2} = 190 \times .12 \times .918 = 0.285$$

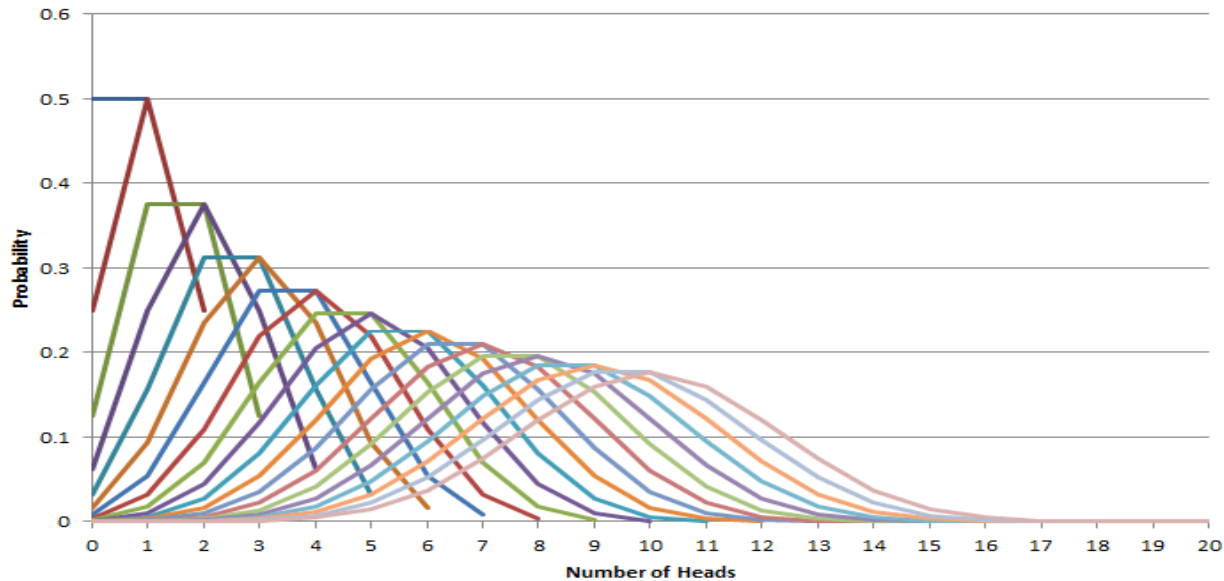
(b) At most 2 invoices will be returned

$$P(r < 3) = P(r = 0) + P(r = 1) + P(r = 2) \\ = 0.1216 + 0.2702 + 0.285 = 0.677$$

	A	B	C
1	=BINOM.DIST(2,20,0.1,FALSE)		
2	0.285179807		
3			
4	=BINOM.DIST(2,20,0.1,TRUE)		
5	0.676926805		

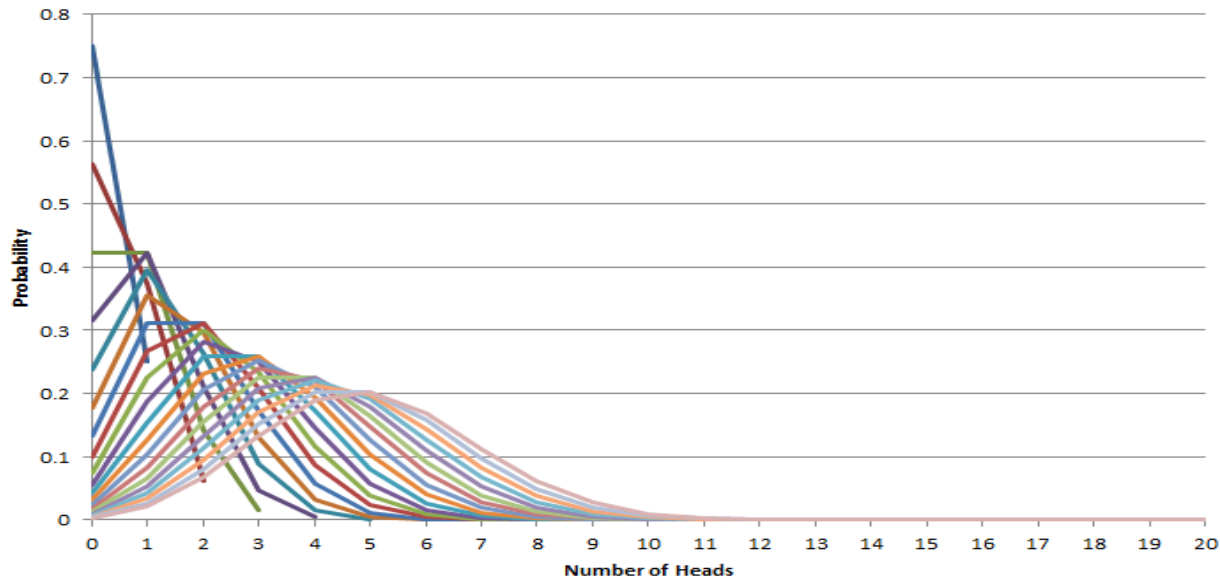
The Binomial Distribution

$(n=1 \text{ to } 20, p=0.5)$



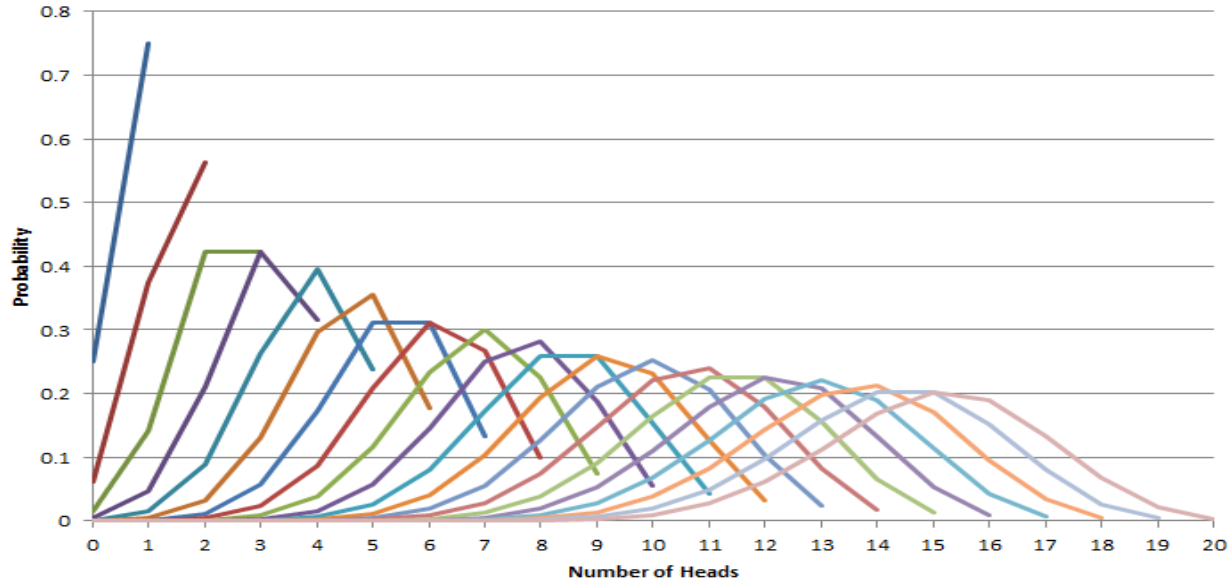
The Binomial Distribution

$(n=1 \text{ to } 20, p=0.25)$



The Binomial Distribution

$(n=1 \text{ to } 20, p=0.75)$



Things to note

The line for $n=1$ stops at 1.

The line for $n=2$ stops at 2.

...

The line for $n=10$ stops at 10.

And so on.

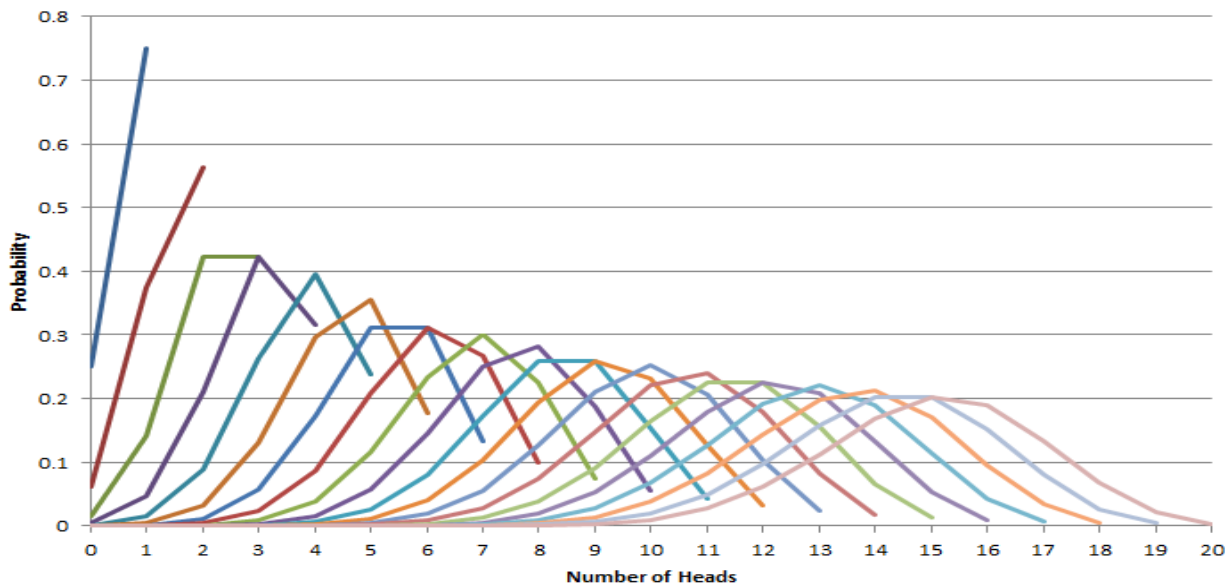
That is, **the Binomial distribution is finite.**

Things to note

- Also, each trial is **mutually exclusive** of the rest, one of the outcomes must happen, and the probabilities add to 1.
- Also, **as n gets large**, the “**main bit**” of the probability distribution **gets smoother**.

Skewness

Note the asymmetry in the previous graphs of the Bin($n=1$ to 20, $p=0.75$) distribution:



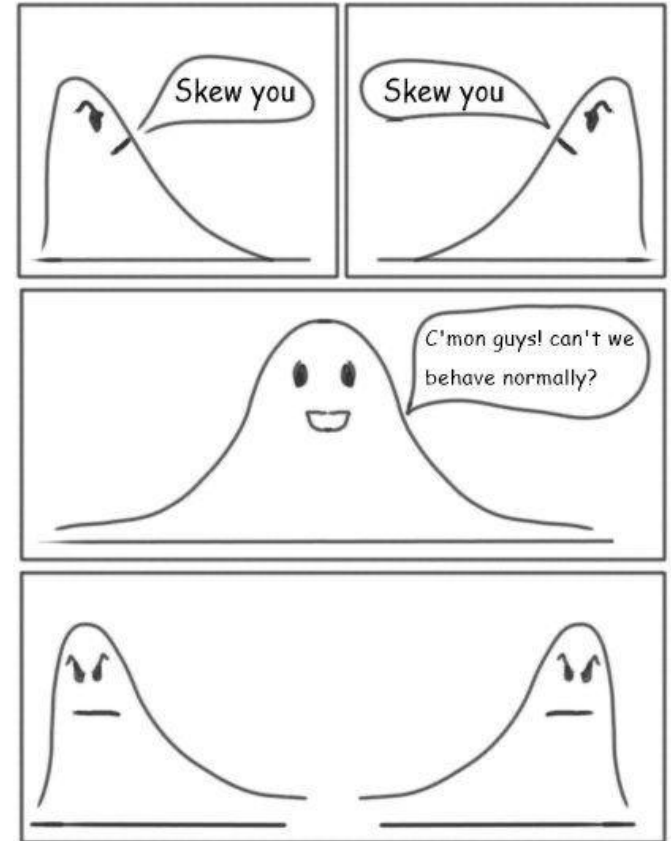
Skewness

As we might expect:

- if **p** is “low” (that is, <0.5), the bulk of the distribution is to the left, leaving a tail on the right.
- If **p** is “high” (>0.5), the bulk is on the right.

This summarises as:

P is	Distribution is
Small (<0.5)	Right-skewed
Exactly 0.5	Symmetric
Large (>0.5)	Left-skewed



An Example

Relevant for what sort of question:

It is known that a manufacturing process produces defective widgets 1% of the time.

These widgets are packed in boxes of 1000.

- a) How many defective widgets would you **expect** in each box?
- b) What is the probability of a box containing precisely **this many** defective widgets?
- c) What is the percentage chance a box has **any** defective widgets?

An Example

Such a question might be:

It is known that a manufacturing process produces defective widgets

1% of the time. p

These widgets are packed in boxes of

1000. n

(a) How many defective widgets would you expect in each box,

mean

(b) what is the probability of a box containing precisely this many (r) defective widgets

$$P(r=?, n=1000, p=0.01)$$

(c) what is the percentage chance a box has any defective widgets?

$$P(r>0, n=1000, p=0.01)$$

An Example

It is known that a manufacturing process produces defective widgets 1% of the time. These widgets are packed in boxes of 1000.

- (a) How many defective widgets would you **expect** in each box?

Expected defective widgets = np

$$= 1000 \times 0.01 = 10.$$

An Example

It is known that a manufacturing process produces defective widgets 1% of the time. These widgets are packed in boxes of 1000.

(b) What is the probability of a box containing precisely **10** defective widgets?

$$P(\text{r successes in n trials}) = {}^nC_r \times p^r \times (1 - p)^{n-r}$$

We already have **n=1000** and **p=0.01**, and we're given **r=10**, so:

$$P(r = 10) = {}^{1000}C_{10} \times 0.01^{10} \times 0.99^{990} = \mathbf{0.126}$$

An Example

It is known that a manufacturing process produces defective widgets 1% of the time.
These widgets are packed in boxes of 1000.

(c) What is the percentage chance a box has **any** defective widgets?

“Any” means “not none”.

For us, **that’s 1 or more.**

i.e. $P(r=1) + P(r=2) + \dots + P(r=1000)$.

That’s a lot of work.

Instead, remember:

$$P(r=0) + P(r=1) + \dots + P(r=1000) = 1$$

So

$$P(r=1) + P(r=2) + \dots + P(r=1000) = 1 - P(r=0)$$

An Example

It is known that a manufacturing process produces defective widgets 1% of the time.
These widgets are packed in boxes of 1000.

(c) What is the percentage chance a box has **any** defective widgets?

“Any” means “not none”.

$$P(r = 0) = {}^{1000}C_0 \times 0.01^0 \times 0.99^{1000} = 0.00004317$$

So

$$\begin{aligned} P(r > 0) &= 1 - P(r=0) \\ &= 1 - 0.00004317 \\ &= 0.99995683 \end{aligned}$$

*In excel =BINOM.DIST(0,1000,0.01,true)

Summary of the session

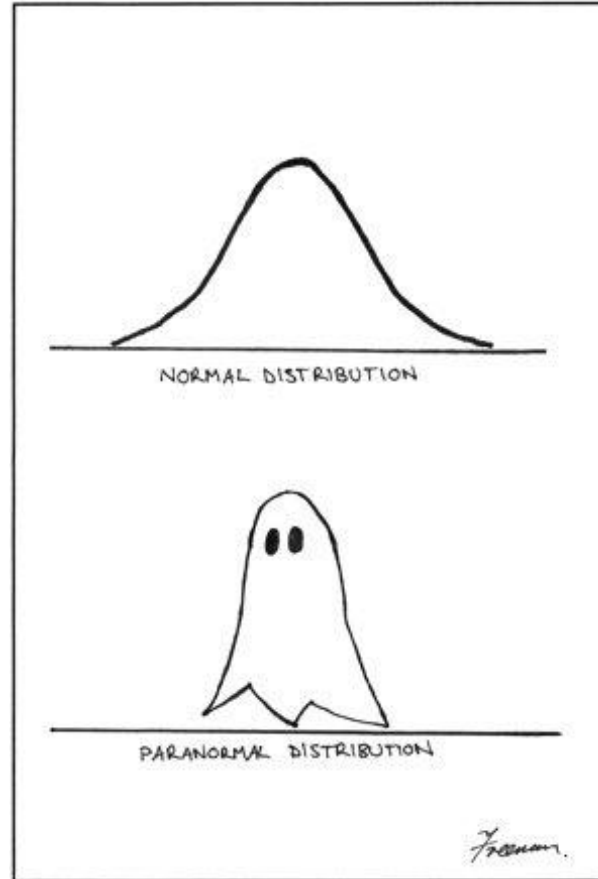


The Binomial Distribution - Is a discrete probability distribution and is appropriate when:

- A variable can **only take on one of two values**
 - The probability of the two outcomes **does not change** from trial to trial
 - Successive trials are **independent**
-
- The **number of ways of choosing r slots for successes out of n trials** *is given by*
$${}^nC_r = \frac{n!}{r!(n-r)!}$$
 - If p is the probability of success, then the probability of r successes out of n trials is
$$P(r \text{ successes in } n \text{ trials}) = {}^nC_r \times p^r \times (1 - p)^{n-r}$$

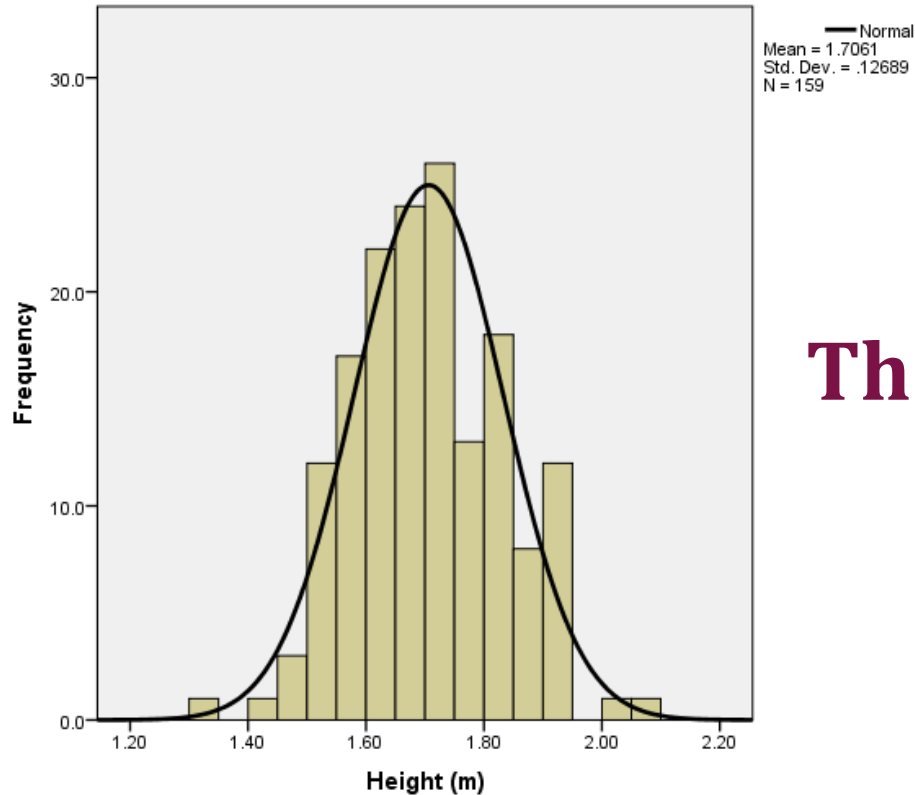
Probability Distributions

(part 2)



Plan of the session

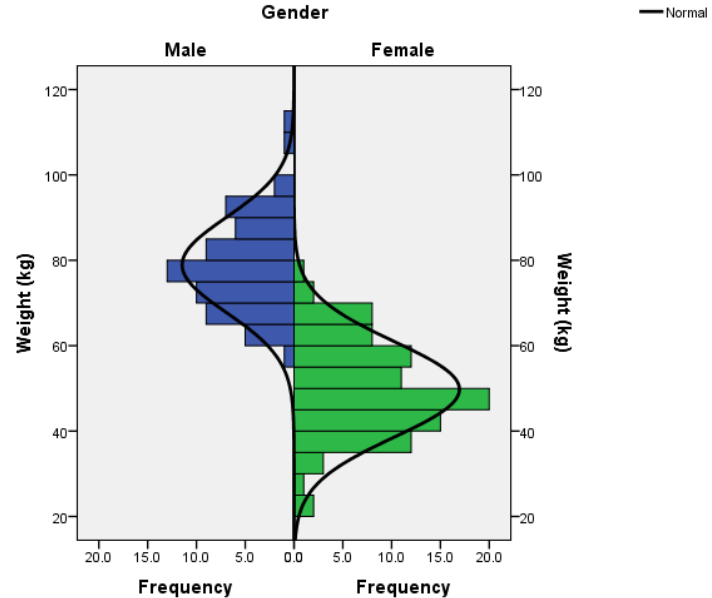
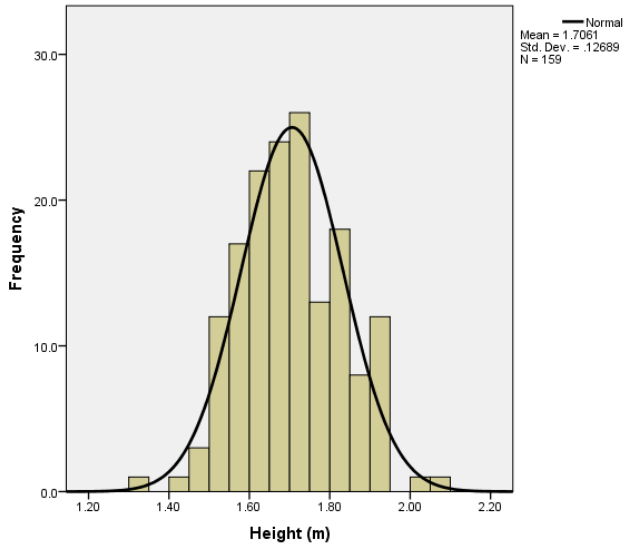
- Probability distributions - what they do and how we use them?
- We looked at:
 - **Bernoulli** ($E(X) = p$ and $\text{Var}(X) = p(1 - p)$)
 - **Binomial** ($E(X) = np$ and $\text{Var}(X) = np(1 - p)$)
- Now we will introduce the
 - **Normal distribution**



The Normal Distribution

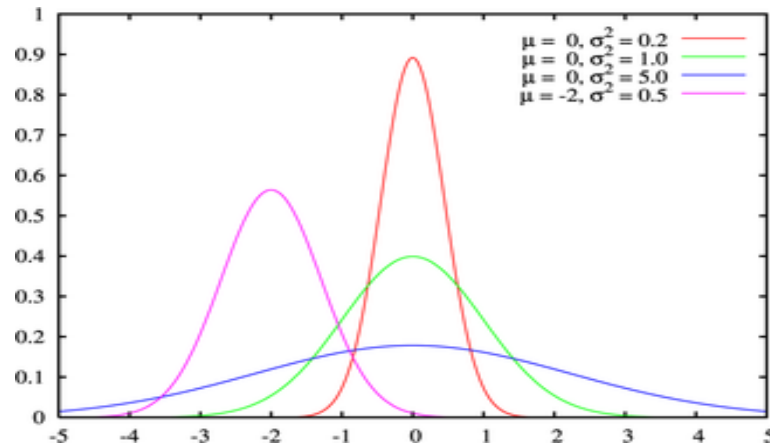
Bell-shaped curve?

Haven't we seen something like the middle of the binomial before?



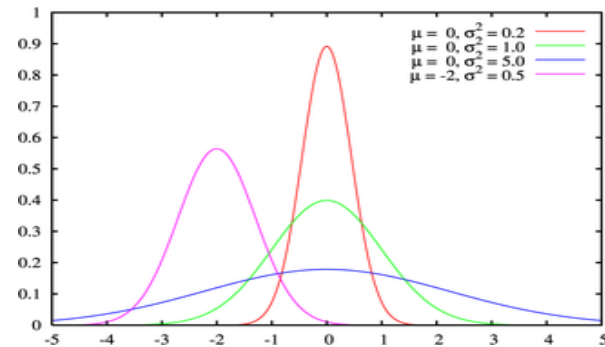
The Normal Distribution

- The **normal distribution** is a common way to model **continuous data**.
- It is a **two parameter** family of distributions given by the **mean μ** (location) and the **variance σ^2** (spread)
- We write **$X \sim N(\mu, \sigma^2)$** to show that X is normally distributed with mean μ (mu) and standard deviation σ (sigma)



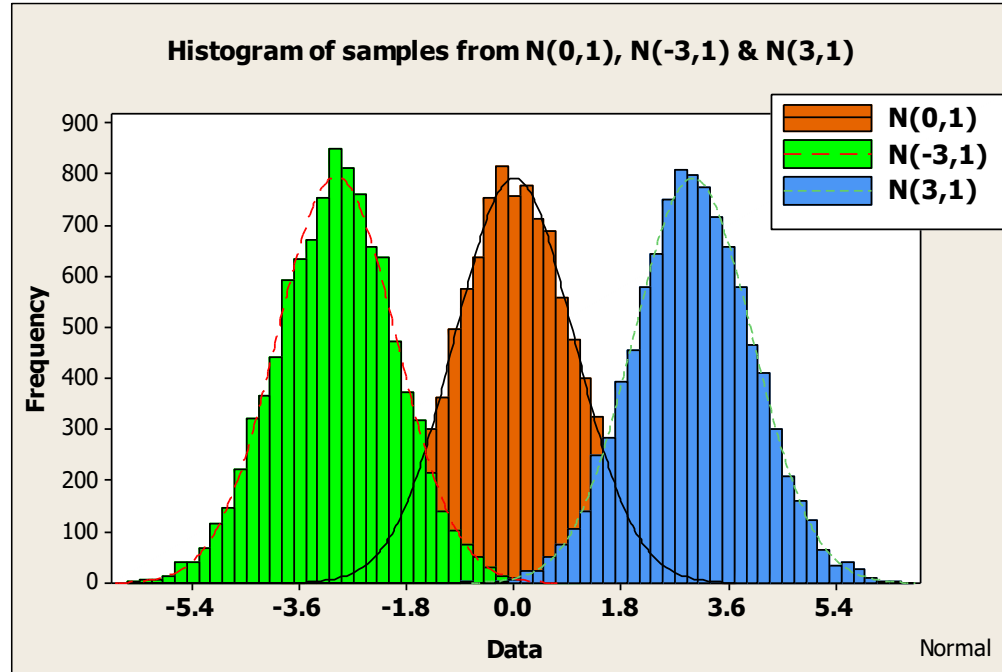
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The Normal Distribution



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What effect does the mean value have on the Normal curve?



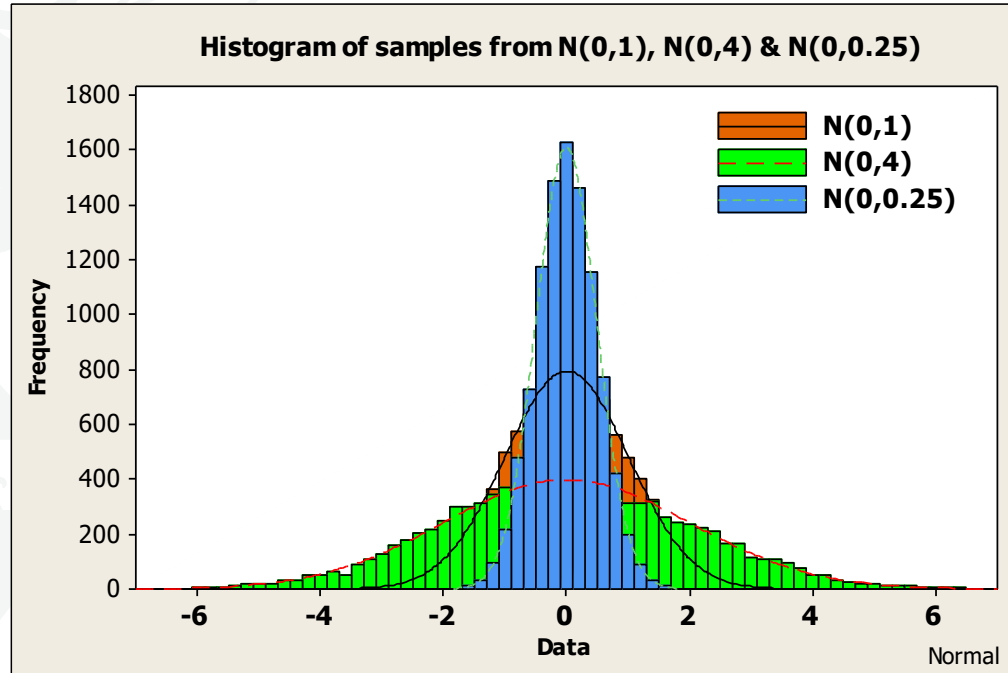
Location

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What happens to the Normal curve as the standard deviation (or variance) changes?

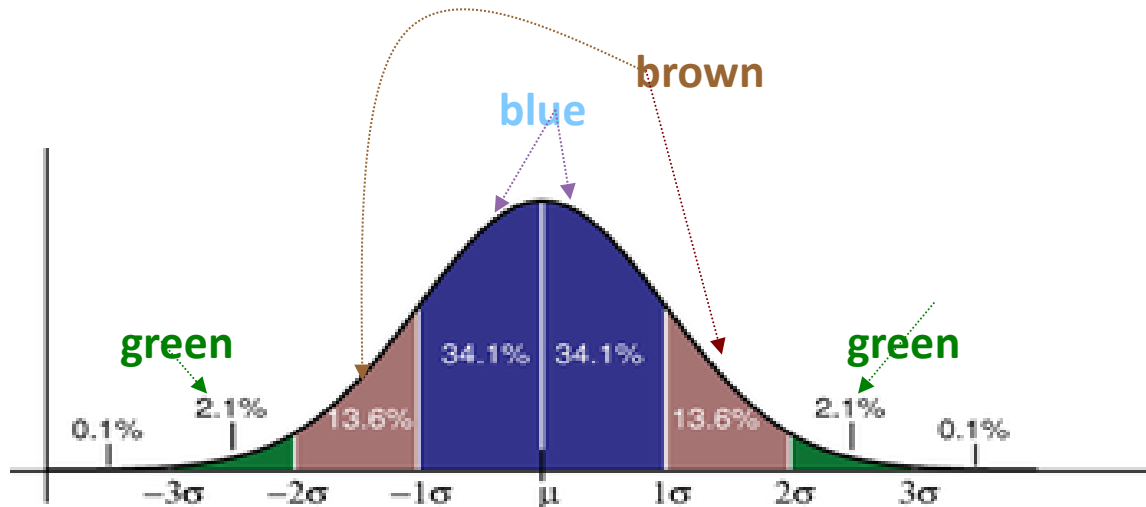
Dispersion



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The Normal Distribution

- **Dark blue** is less than one standard deviation from the mean. For the normal distribution, this accounts for about **68%** of the set while two standard deviations from the mean (blue and brown) account for about **95%** and three standard deviations (blue, brown and green) account for about **99.7%**.



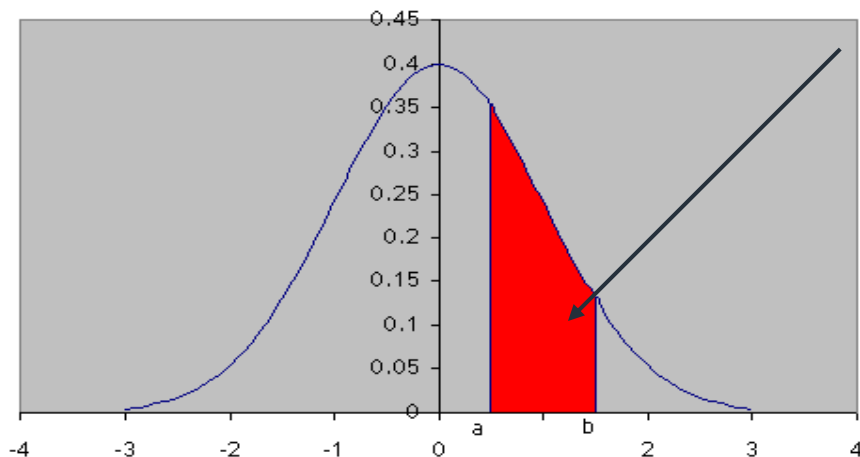
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The Normal Distribution cont.

The area under the curve between two points on the horizontal axis represents the probability that the value of the variable lies between these two points.

$P(a < x < b)$ = area under the curve between points a and b.

Probability Density function

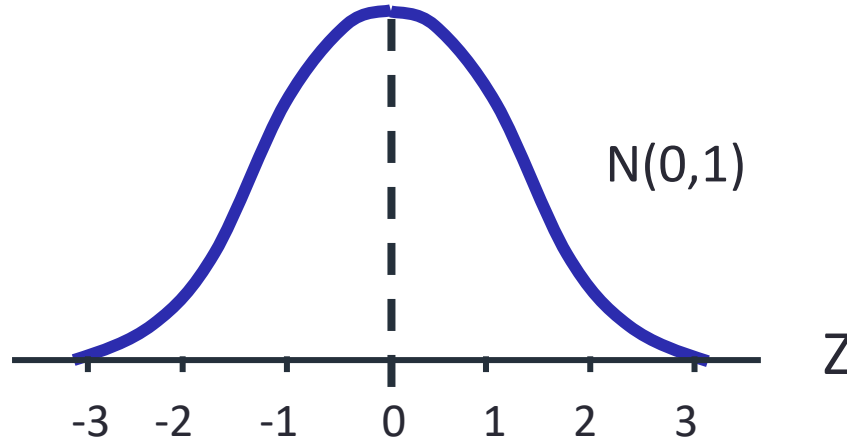


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The Standard Normal Curve credited to *Karl Gauss* (1777 – 1855)

Mean = 0

Variance = 1



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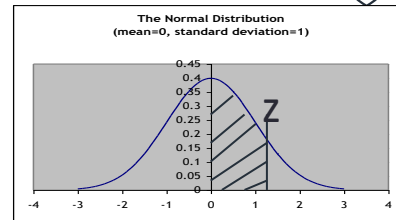
- The numbers on the horizontal axis = the number of standard deviations from the mean & are called **z values**.
- **z-tables (normal tables)** exist that list the areas under this curve for certain ranges of z .

A note about the Tables in Books...

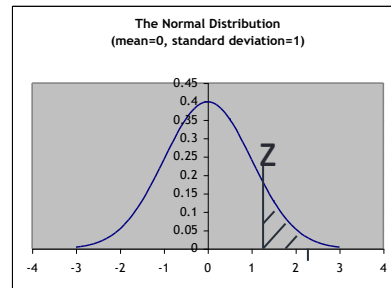


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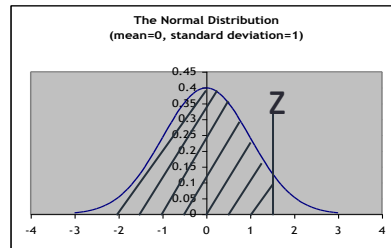
- Some books give you the table for the probabilities for the area between the mean and the z values.



- Some give you the probabilities for the area of the tail from the z value.



- Some give you the probabilities of the area below the z value.

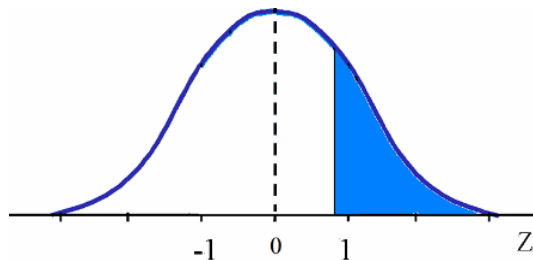


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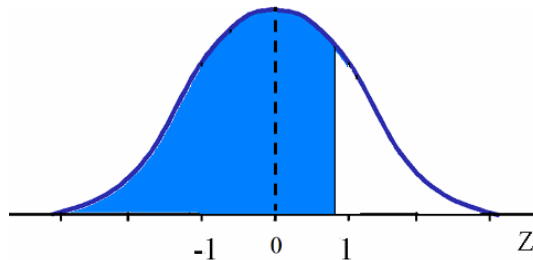
How do we read tables?

- Using the table:

Find $P(z \geq 0.82)$



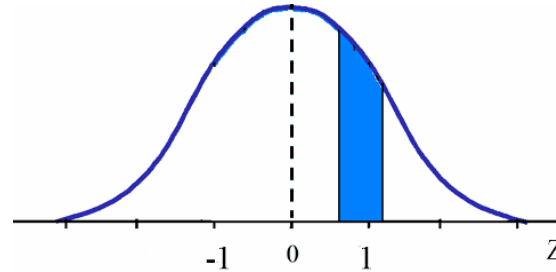
Find $P(z \leq 0.82)$



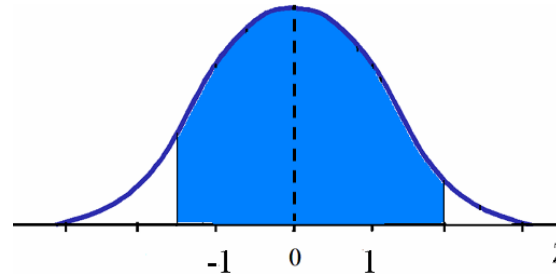
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How do we read tables? cont.

Find $P(0.65 \leq z \leq 1.22)$

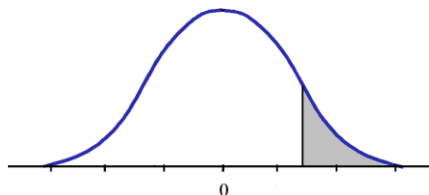


Find $P(-1.5 \leq z \leq 2)$



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How do we read tables? cont.



Area given in the table

Z	0.00	0.01	0.02	0.03	0.04	0.05
0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578
0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977
1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056
1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606
2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202

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Standardising Normal Distributions



- **Normal z-table** can only be used for the standard **Normal distribution $N(0,1)$** .
- To solve problems for Normal distributions with mean values $\neq 0$ and standard deviations $\neq 1$, need to use the **Z transformation**.
- This calculates the number of standard deviations from the mean your value of interest (x) is

Z transformation

$$Z = \frac{\text{value of interest} - \text{mean}}{\text{standard deviation}}$$

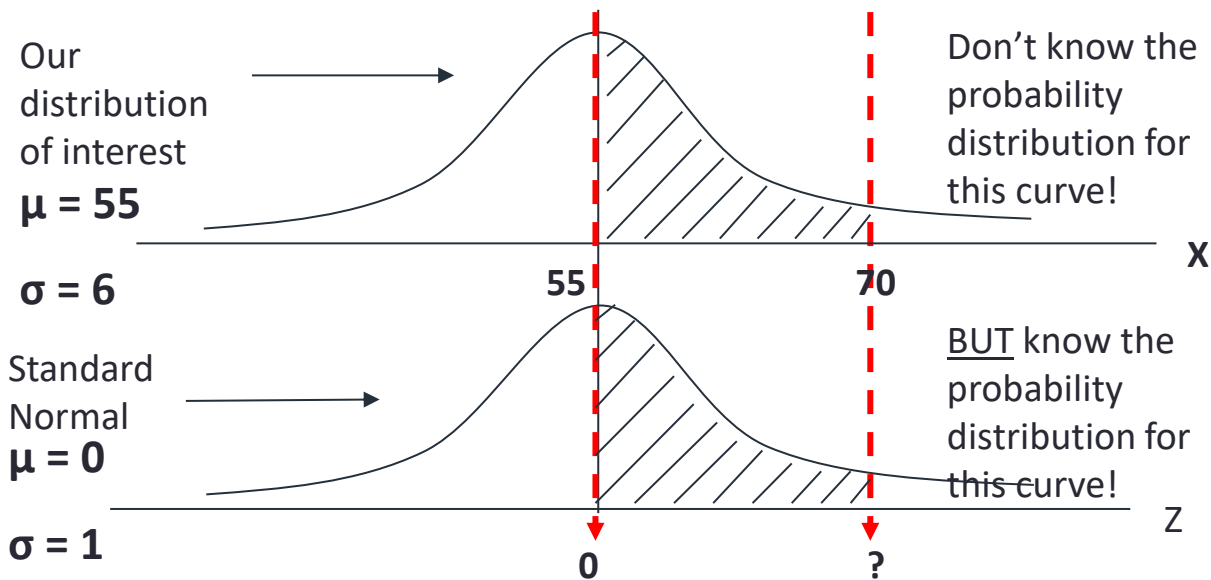
$$Z = \frac{x - \mu}{\sigma}$$

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Example

200 pupils sit a maths test. The mean score is 55% with a standard deviation of 6%.

What is the probability that a randomly selected score is between 70% and the mean?

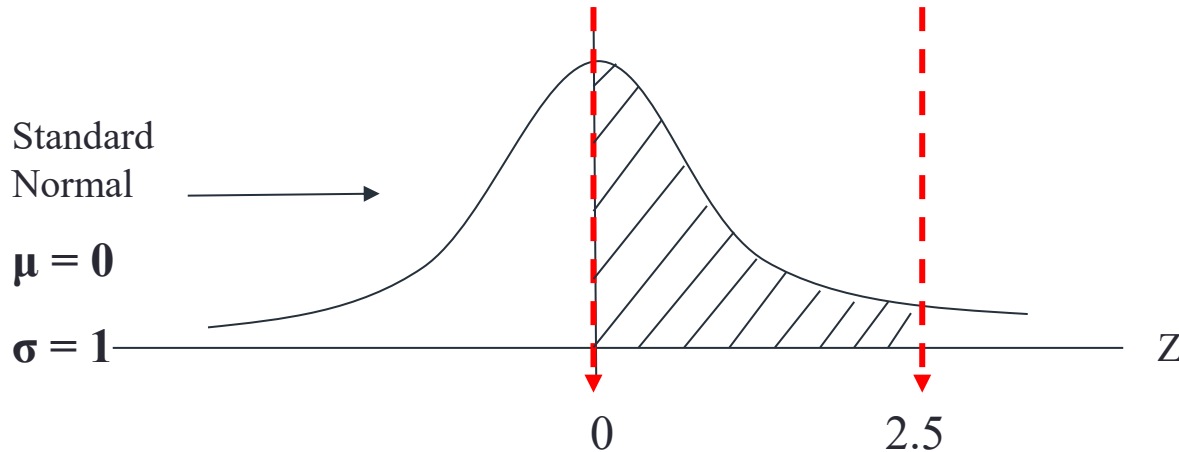


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Use the z-transformation

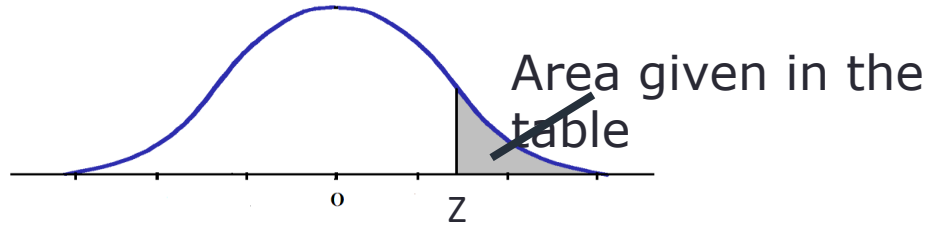
$$z = \frac{x - \mu}{\sigma}$$

$$z = \frac{70 - 55}{6} = 2.5$$



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Use the z-transformation cont.



Z	0.00	0.01	0.02	0.03	0.04
0.0	0.5000	0.4960	0.4920	0.4880	0.4840
2.5	0.0062	0.0060	0.0059	0.0057	0.0055

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$$\text{So, } P(55 \leq x \leq 70) = 0.5 - 0.0062 = 0.4938$$

Example

A batch of loaves is baked. The weight of loaves is normally distributed with $\mu = 800\text{g}$ & $\sigma = 10\text{g}$.
What proportion will weigh between 805g & 815g ?

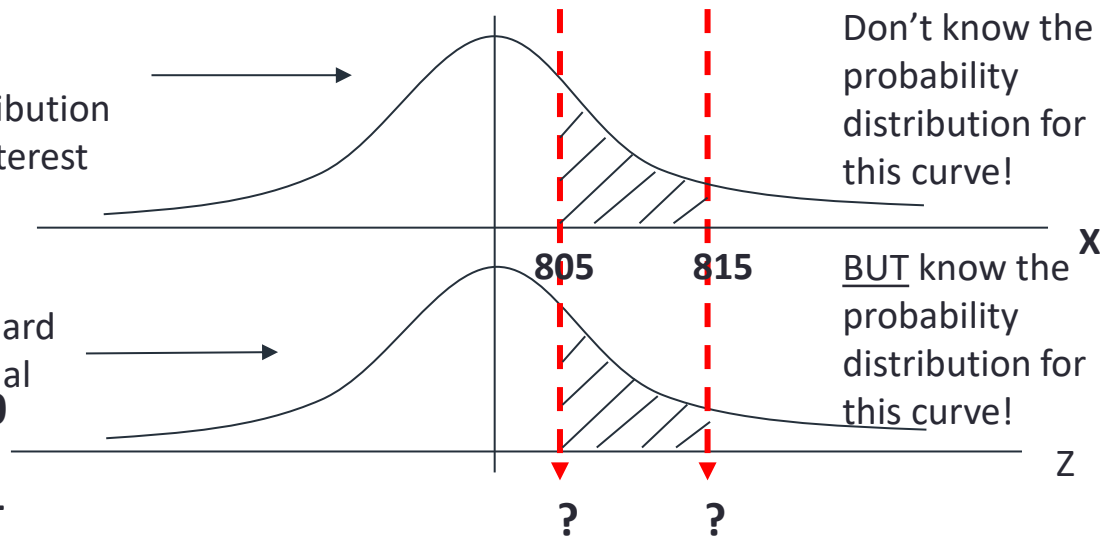
$\mu = 800$

$\sigma = 10$

Our
distribution
of interest

Standard
Normal
 $\mu = 0$

$\sigma = 1$

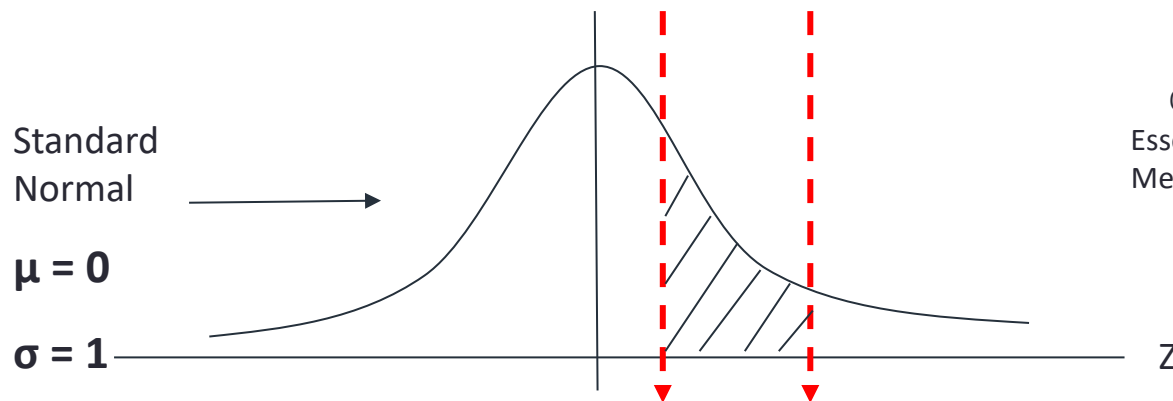


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Use the z-transformation

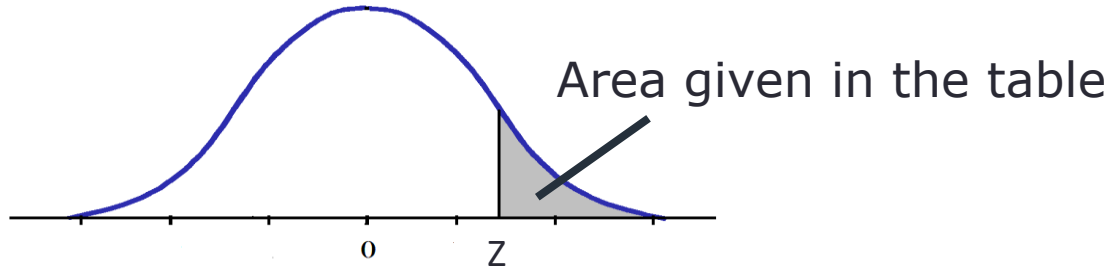
$$z = \frac{x - \mu}{\sigma}$$

$$z = \frac{805 - 800}{10} = 0.5, \quad z = \frac{815 - 800}{10} = 1.5$$



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Use the z-transformation cont.



Z	0.00	0.01	0.02	0.03	0.04
0.5	0.3085	0.3050	0.3015	0.2981	0.2946
1.5	0.0668	0.0655	0.0643	0.0630	0.0618

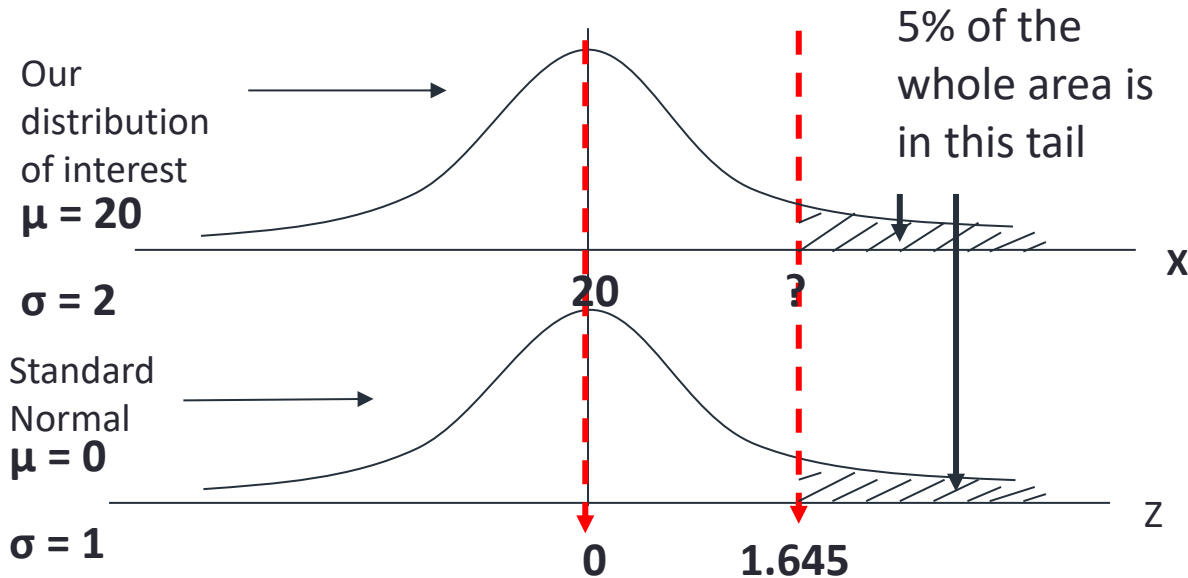
$$\text{So, } P(805 \leq x \leq 815) = 0.3085 - 0.0668 \\ = 0.2417,$$

so 24.17% of loaves weigh between 805g & 815g.

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Example

A company pays its employees an average wage of £20 per hour, with a standard deviation of £2. If the wages are normally distributed, the highest 5% of the hourly wages is greater than what amount?



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Use the z-transformation

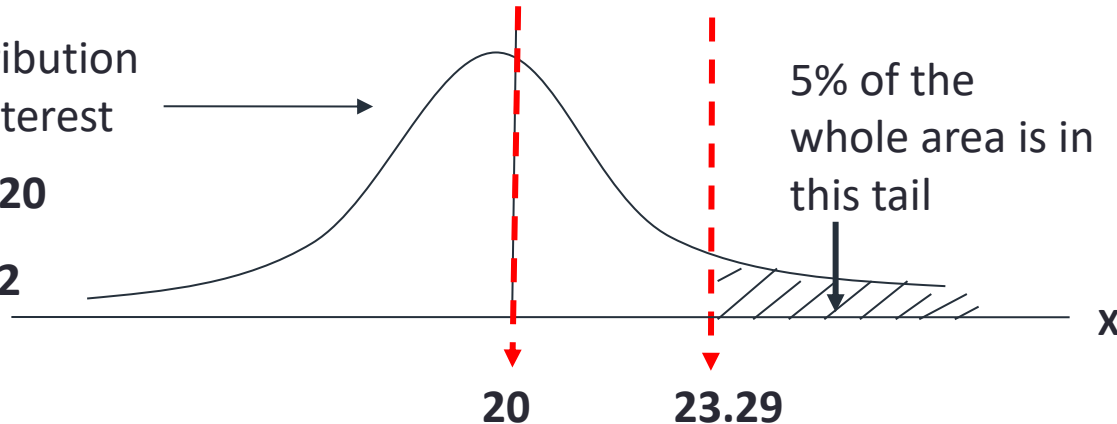
$$z = \frac{x - \mu}{\sigma}, \quad 1.645 = \frac{x - 20}{2},$$

$$x = (2 \times 1.645) + 20 = \text{£}23.29$$

Our
distribution
of interest

$$\mu = 20$$

$$\sigma = 2$$



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Finally

- We've seen as n gets large, the binomial looks more like (it approximates) the normal distribution. We can use that property to answer questions that would otherwise be difficult.
- **For example:** York has a population of **198,051**. **12%** of people have O-blood. What's the probability that there were more than **24,000** people with O- blood?

Finally

- This appears to be a **binomial** question. We have **n=198,051** and **p=0.12**.

(there are independence assumptions being made here)

— We want **the probability of more than 24,000**.

That's

$$P(r = 24000) + P(r = 24001) + P(r = 24002) + \dots + P(r = 198051)$$

Or

$$1 - (P(r = 0) + P(r = 1) + \dots + P(r = 23999))$$

Those calculations would both be very time consuming.

Finally

We can use the fact that the **binomial approximates the normal distribution** to save time.

We **expect** $198,051 \times 0.12 = 23,766.12$ people to have type O- blood.

That's our mean, μ .

We can calculate

$$\sigma = \sqrt{n \times p \times (1 - p)} = \sqrt{198051 \times 0.12 \times (1 - 0.12)} = 144.62$$

Finally

Now we just need **z**.

$x = 24000$, so

$$z = \frac{x - \mu}{\sigma} = \frac{24000 - 23766.12}{144.62} = 1.62$$

The table gives (*row 1.60, column 0.02 of the tables uploaded on VLE*) 0.94738.

This is the probability of **less than** 24,000.

The probability of **more than** 24,000 is

$$1 - 0.94738 = 0.05262$$

or about 5% of York's population .

Summary

- We have looked at probability distributions.
- What they do and how we use them.
- We looked at:
 - Bernoulli
 - Binomial
 - Normal
- These neatly match up with dichotomous and continuous data and lead nicely into future topics.

Next topic

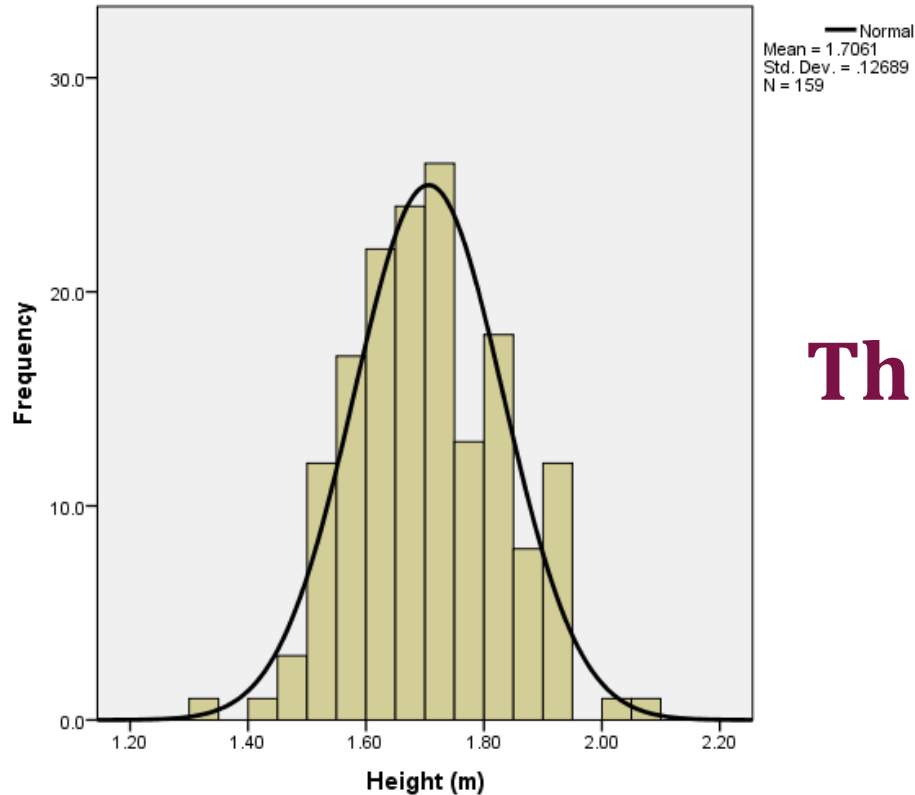
Data Sampling

Reading: Oakshott's chapters 3 and 8

- **Chapter 3** Surveys and Samples
- **Chapter 8** Analysis of Sample Data

Last session

- We have looked at probability distributions.
- What they do and how we use them.
- We looked at:
 - Bernoulli
 - Binomial
 - Normal
- These neatly match up with dichotomous and continuous data and lead nicely into future topics.



The Normal Distribution

Some more examples

Issues with raw data

- “What’s the probability someone is at least 1.84m tall?”

We already know

$$\mu=1.7061m \text{ and } \sigma=0.12689m.$$

? We need to do is convert what we **have**, to what we **know about** ...

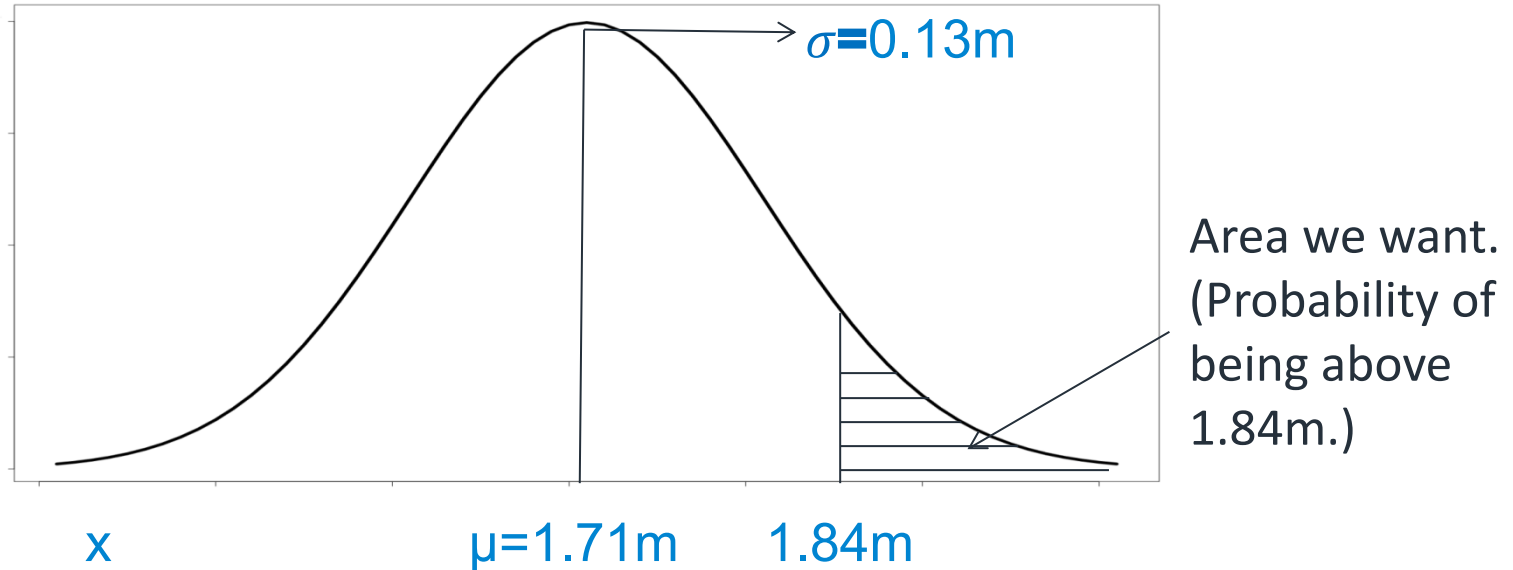
There are infinitely many normal distributions. We can’t have a conversion table for each!

The Standard Normal

- The conversion table we want is for the **Standard Normal**.
- That is, for the **Normal Distribution** with **mean of 0** and **standard deviation of 1**.

An Example

It is always good to draw a diagram to help clarify.



Converting to the Standard Normal

Converting to the standard normal is managed by the formula:

$$z = (x - \mu) / \sigma$$

where x is the data point in question.

Converting to the Standard Normal

For this example, $x=1.84m$.

That means

$$z=(x-\mu)/\sigma=(1.84m-1.71m)/0.13m=1.00$$

- i.e. the **data point 1.84m** (don't forget the units for the final answer) **becomes 1** on the **standard normal distribution**.

Interpretation of z

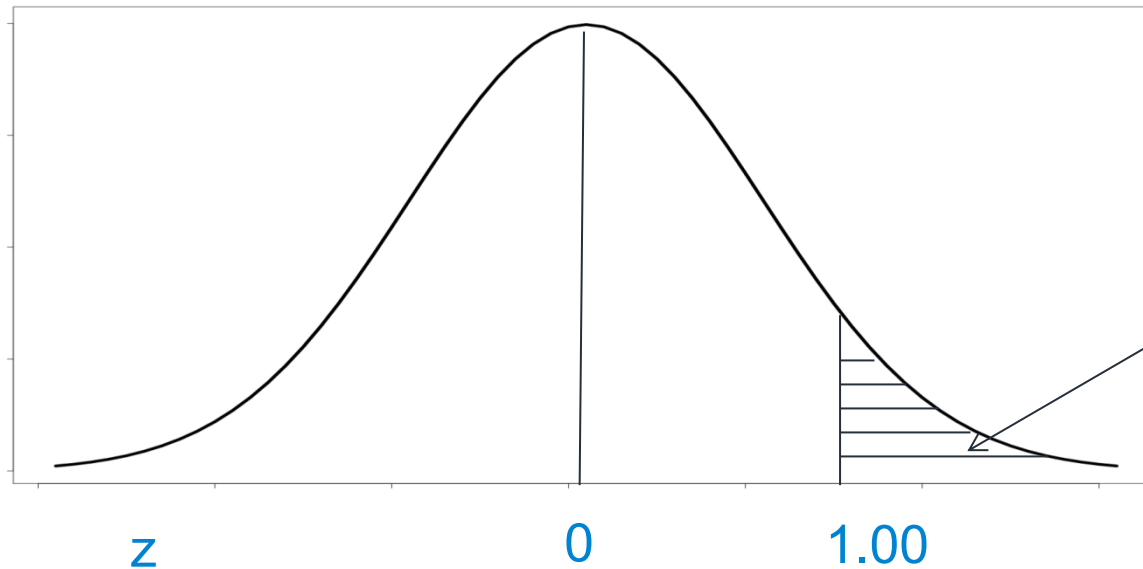
Recall, **z** can be thought of as “**the number of standard deviations x is above (or below) the mean**”.

In this case, **1.84m is one standard deviation (0.13m) above the mean (1.71m).**

(i.e. $1.84m = 1.71m + 1 \times 0.13m$)

An Example

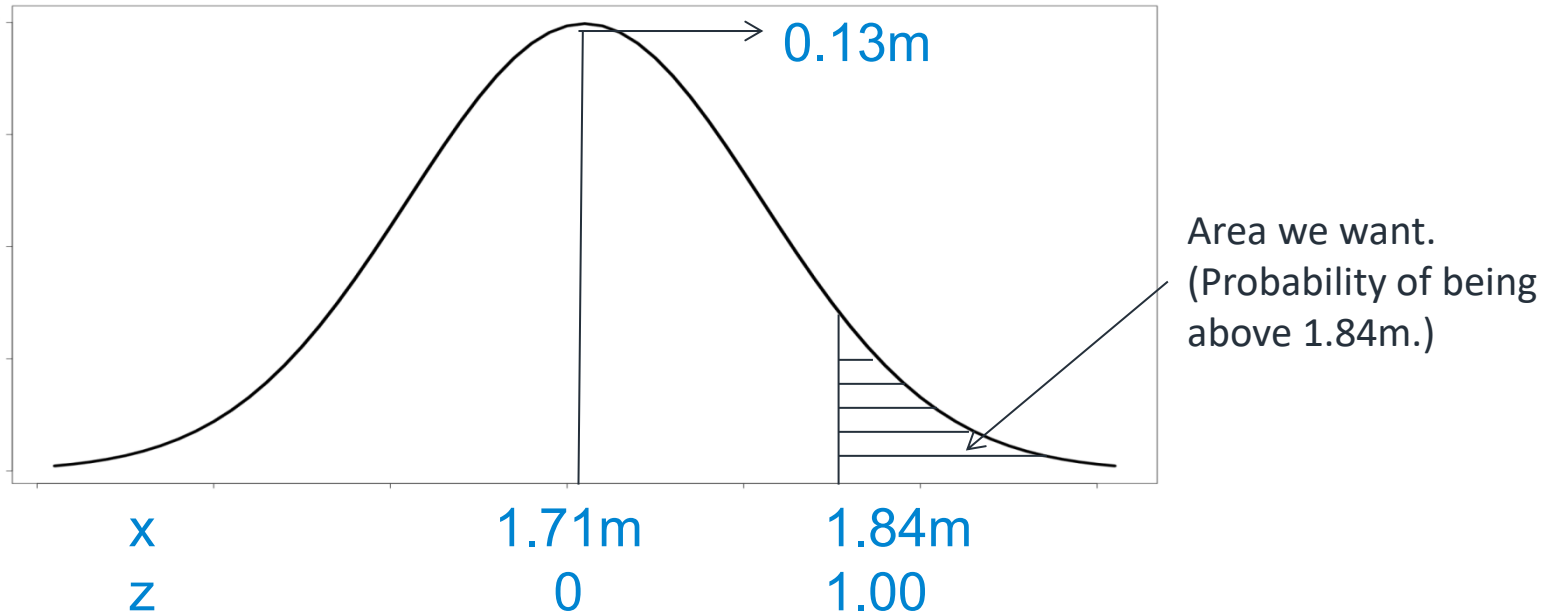
If we go back to the diagram, what we have is



area we want.
(Probability of
being above
1.84m.)

An Example

Lets put both rows on at once below the horizontal axis:



An Example

- We want to be looking at the **standard normal table** (see handout entitled **Normal Distribution Tables on the VLE**).
- The **left-hand column** is for the first two digits of z .
- The **top most row** is to specify the third digit.

An Example – using the table

- The **left-hand column** is for the first two digits of z .
- The **top most row** is to specify the third digit.
- So a **value for $z=1.23$** would be found on row **1.20** and column **0.03**.

(0.89065)

- And a **value for $z=-1.87$** would be found on row **-1.80** and column **0.07**.

(0.03074)

Back to the Example

Our example: look at the standard normal table on the side that has 0.00 as the top entry at the left.

Look down the left column until you see 1.00 and take the first column (0.00).

You **should see 0.84134**; this is **the entry for 1.00**.

This is the area of the normal distribution below $z=1.00$. That is, the bit to the **left** of the line.

Back to the Example

We know the **area to the left of $z = 1$** is 0.84134, so the area **to the right** must be:

$$1 - 0.84134 = 0.15866$$

That implies the probability of someone being above 1.84m tall is about 0.16, or about 1 in 6.

Another Example

If we'd been looking for the probability of someone being under 1.63m, we'd find $z = (1.63 - 1.71) / 0.13 = -0.62$. Flip the page over, and now we want the -0.60 row and 0.02 column, for a value of **0.26763**.

We don't have to do the "1 minus" step in this case, as we are only looking for an area to the left, which is what the table provides directly.

From z to x

$$z = (x - \mu) / \sigma$$

If we know μ and σ , we can calculate x from z .

$$z = \frac{x - \mu}{\sigma}$$

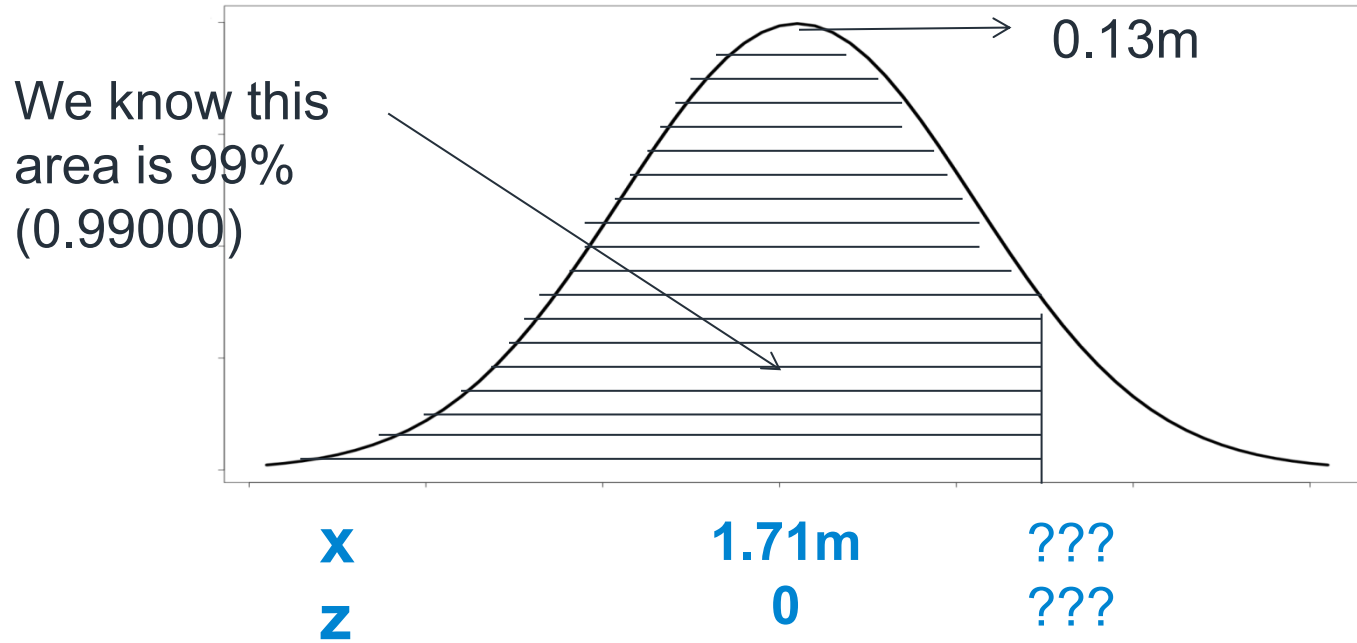
If we know μ and σ , we can calculate x from z .

$$x = \mu + z\sigma$$

This lets us work the other direction.

A More Difficult Example

How tall are the tallest 1% of people? Give the shortest height



A More Difficult Example

- We need to find an entry in the table as close to **99% (0.99000)** as we can.
- The best is for $z = 2.33$ (the entry is 0.99010).
- This implies x must be
- $x = \mu + z\sigma = 1.71m + 2.33 \times 0.13m = 2.01m$
(i.e. 6'7")

A More Difficult Example

- If two entries are equally close, which should you use?

Well, that depends.

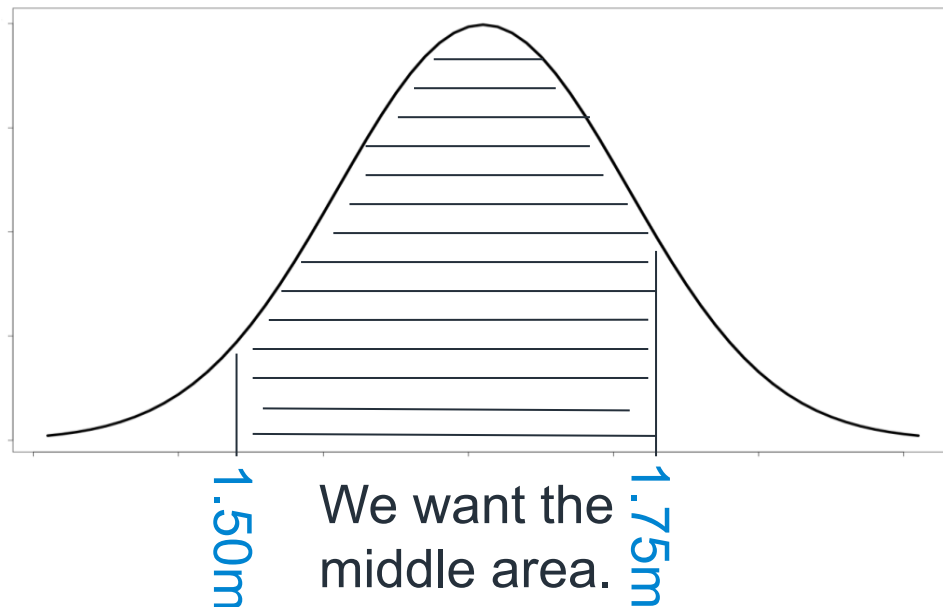
- We wanted the shortest height of the tallest people.

$z=2.32$ would have given us the tallest height of the shortest people.

In each question, we have to think which answer is better.

Two bounds

Someone with a liking for round numbers wonders how many people are between 1.50m and 1.75m tall. Once again, as always, a diagram is helpful:



Two bounds

- 1.75m is

$z = (1.75 - 1.71) / 0.13 = \mathbf{0.31}$ standard deviations above the mean.

The table gives us **0.62172**. This is the area under the curve to the left of 1.75m.

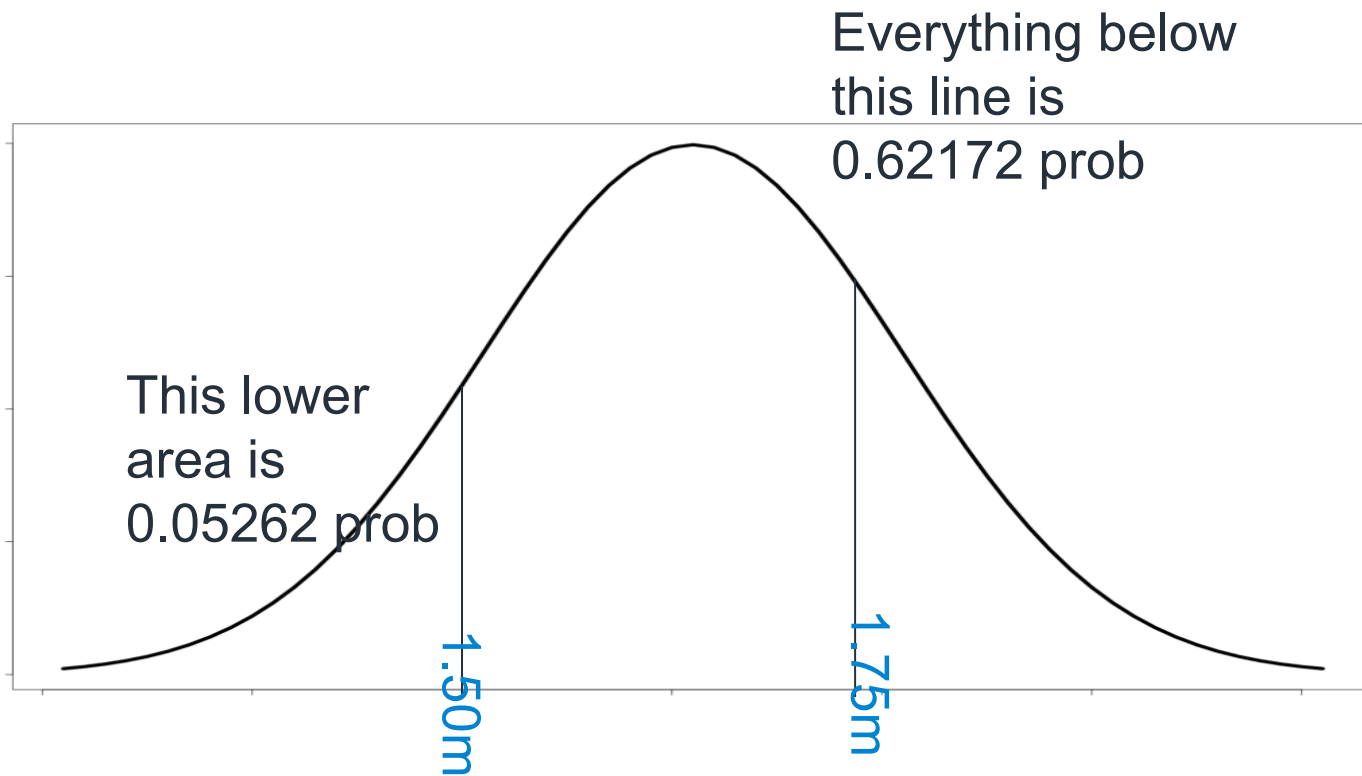
- 1.50m gives

$z = (1.50 - 1.71) / 0.13 = \mathbf{-1.62}$.

That is, it's 1.62 standard deviations below the mean.

The table gives **0.05262**.

Two bounds



Two bounds

The area we want is the area below the line on the right minus the area below the line on the left. That is,

$$0.62172 - 0.05262 = \mathbf{0.56910}.$$

The probability of someone being between these heights is 0.57.

Hence, 57% of the population are between 1.50m and 1.75m.

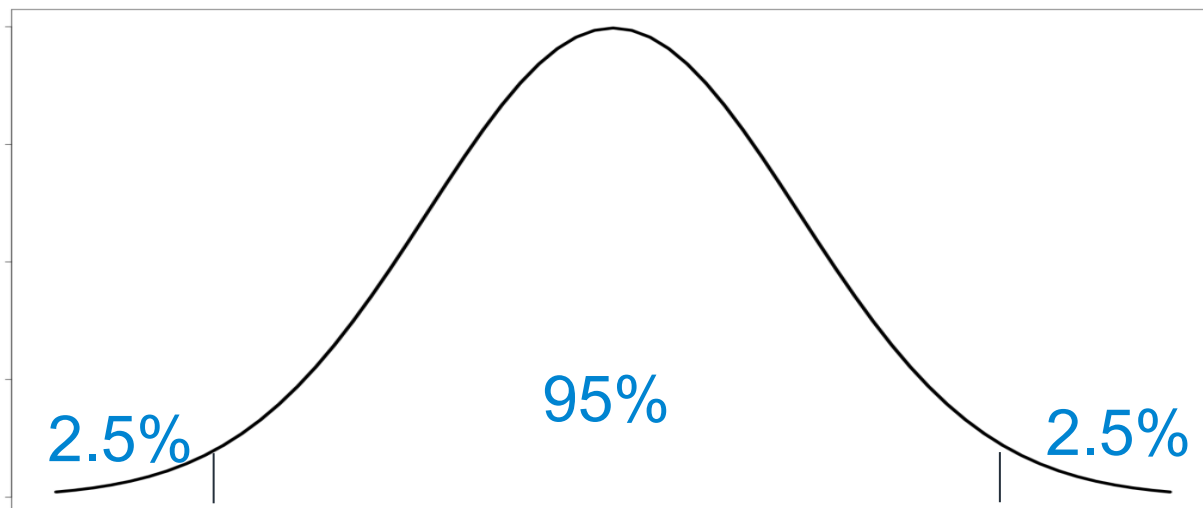
Prediction Intervals

- We can use these ideas to predict where we expect data to be.
- That is, to create a “**prediction interval**”, two numbers that surround 90%, 95%, 99% etc. of the data.
- This is making the “**general layout**” a little more precise.
- We’re also only going to be looking at **symmetric intervals** laying around the mean.

Prediction Intervals

This is making the “**general layout**” a little more precise.

We’re also only going to be looking at **symmetric intervals** laying around the mean.



Prediction Intervals

- Find that $z=\pm 1.96$.
- That is, $z=1.96$ for the bigger number, and $z=-1.96$ for the lower.
- And this is always true, for any dataset!

Prediction Intervals

So, for the height data, the 95% prediction interval is:

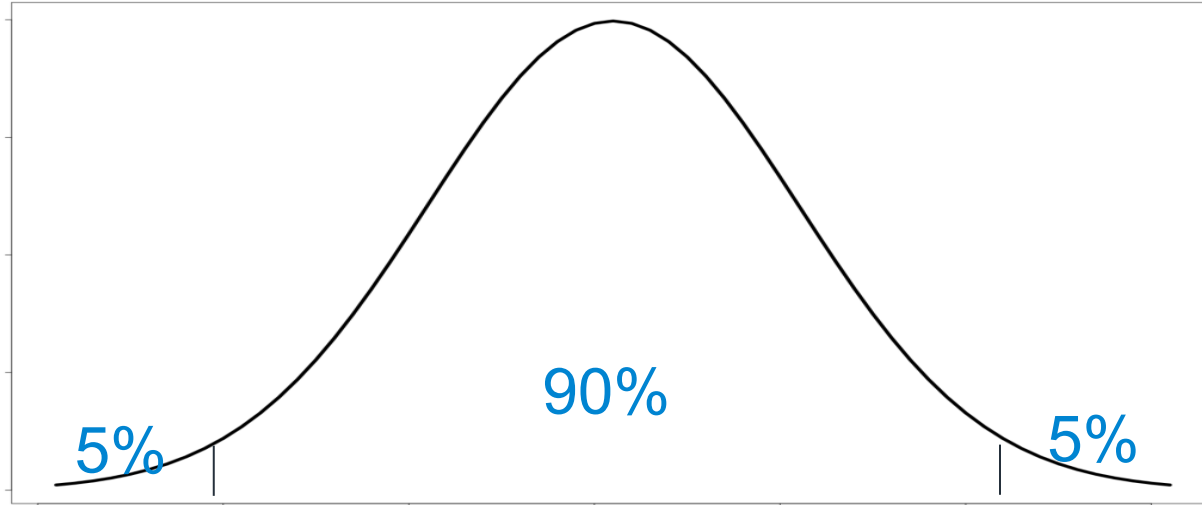
$$\mu \pm z \times \sigma$$

$$1.71\text{m} \pm 1.96 \times 0.13\text{m}$$

$$1.45\text{m to } 1.96\text{m}$$

We would expect 95% of the people to have heights between these bounds.

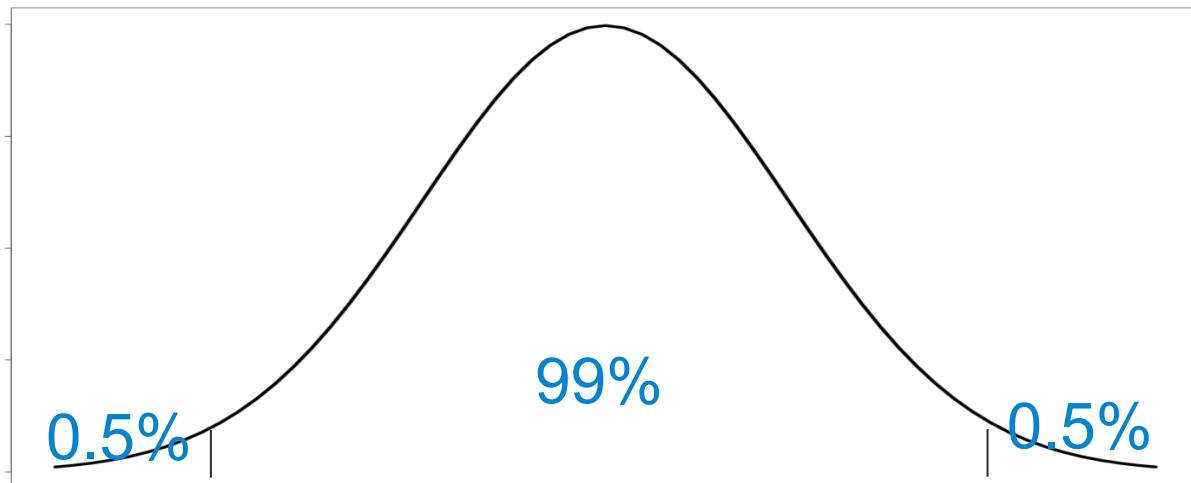
Prediction Intervals



For a 90% prediction interval, we want 5% in each tail:

and we find $z = \pm 1.65$. (1.64 would be slightly too small, that would only be a 89.9% prediction interval.)

Prediction Intervals



For a **99% prediction interval**, we want **0.5%** in each tail and we find **$z=\pm 2.58$** . (2.57 would be slightly too small, that would only be a 98.98% prediction interval.)

Next topic

Data Sampling

Reading: Oakshott's chapters 3 and 8

- **Chapter 3** Surveys and Samples
- **Chapter 8** Analysis of Sample Data